

CNW 2140 Computer Networks

Course syllabus

School of Computing, Tarnhollow University

Autumn term 2026. Version 1.1, issued Tuesday 1 September 2026.

1. Teaching team

Lecturer: Dr. Oleska Brantigern, room HV 3.07. Office hours are Wednesday 14:00 to 15:30, no appointment needed.

Teaching assistant: Juno Feathersmythe, who runs the labs and marks the lab reports. Lab drop-in is Monday 16:00 to 17:00 in room HV 1.22.

Use the course forum for anything other students might also want to know. Email is for personal matters only, and is read within two working days.

2. Meetings

Lectures are on Tuesday and Thursday, 10:00 to 11:30, in the Hale-Voss Building, room HV 2.14. The lab session is Friday 13:00 to 15:00 in HV 1.22. There are 22 lectures. Teaching starts on Tuesday 8 September 2026 and the last lecture is on Thursday 26 November 2026.

3. What the course covers

The course follows a packet from the wire to the application and back: how hosts are named and found, how the web's protocol is built on a reliable transport, how that transport shares a congested path, and how packets are addressed and routed. The first half of the term is planned as follows.

Lecture	Date	Topic
1	Tuesday 8 September	The layered model and what a packet is
2	Thursday 10 September	Links, frames and error detection
3	Tuesday 15 September	Switching and Ethernet
4	Thursday 17 September	Sockets and the transport layer
5	Tuesday 22 September	DNS
6	Thursday 24 September	HTTP
7	Tuesday 29 September	TCP congestion control
8	Thursday 1 October	TCP reliability and retransmission
9	Tuesday 6 October	IP addressing and subnets
10	Thursday 8 October	Routing inside a network
11	Tuesday 13 October	Routing between networks
12	Thursday 15 October	TLS, QUIC and HTTP/3

Tuesday 20 October is a revision session with no new material. The second half of the term (lectures 13 to 22) covers wireless networks, network security and data-centre networking; the detailed plan is published after the midterm.

4. Assessment

- Labs: 20% of the course mark. Five labs, 4% each.
- Midterm exam: 25% of the course mark. Thursday 22 October 2026, in the lecture slot, 80 minutes. Covers lectures 1 to 8.
- Group project: 15% of the course mark. Groups of three. Report due Friday 27 November 2026.
- Final exam: 40% of the course mark. Tuesday 15 December 2026, 09:00 to 12:00, Sports Hall B. Covers the whole course.

To pass the course you need an overall mark of at least 50 and at least 45 out of 100 in the final exam.

Materials: the midterm is closed book. For the final exam you may bring one A4 sheet, handwritten on both sides. Calculators are not needed and not allowed.

5. Labs and deadlines

Lab reports are submitted through the course site by 17:00 on the due date.

- Lab 1, packet capture warm-up: due Friday 18 September 2026
- Lab 2, a socket echo server: due Friday 2 October 2026
- Lab 3, TCP trace analysis: due Friday 16 October 2026
- Lab 4, build a DNS resolver: due Friday 6 November 2026
- Lab 5, an HTTP cache: due Friday 20 November 2026

Late policy: a late lab report loses 10 marks out of 100 for each day or part of a day after the deadline, for up to 3 days. After 3 days it scores zero. Extensions are granted only through the School office, not by the lecturer or the teaching assistant.

Regrade requests must reach the teaching assistant within 7 days of the mark being released.

6. Reading

The course text is Hallway and Minch, *Networks from the Wire Up*, 3rd edition. Section numbers on the slides refer to this edition. The library holds twelve copies on short loan, and the e-book is available through the library catalogue.

7. Working together and working honestly

Lab reports are individual work. You may discuss approaches with other students, but the captures, the code and the write-up must be your own. The group project is marked as a group, with a short individual reflection from each member. Suspected plagiarism is referred to the School's integrity officer, and the lecturer has no discretion in that.

If you have a learning support plan, send it to the lecturer in the first two weeks of term so that exam arrangements can be made in time.